MIL-C-46460A (Ord)
26 June 1962
SUPERSEDING
MIL-D-46460 (Ord)
19 December 1960

MILITARY SPECIFICATION

CHARGE, DEMOLITION, BLOCK, M91 PARTS FOR LOADING, ASSEMBLING, AND PACKING

1. SCOPE

1.1 This specification covers special requirements not covered by the drawings and quality assurance provisions for one type of demolition item designated as Charge, Demolition, Block, M91.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-I-45208 - Inspection Requirements, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.

DRAWINGS

ORDNANCE CORPS

8826645 - Charge, Demolition, Block, M91, Assembly

FSC: 1375

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8798858 - Carton, Packing, Ammunition, For Charge, Demolition, Block, M91

8798859 - Box, Packing, Ammunition, For Charge, Demolition, Block, M91.

PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.
IEL-8826645 - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Assembly.-The assembly shall comply with all requirements specified on Drawing (dwg.) 882665 and with all requirements specified in applicable specifications.

3.3 Magnet strength.-The magnets of the complete assembly shall not release the assembly from a flat steel surface approximately 1/8 inches thick when 15 pounds, minimum, is applied normal to the plate; the force being applied equally to each of the four magnets, nor shall it release from a curved steel plate approximately 1/8 inch thick of 5 inch radius when 8 pounds, minimum, is applied in the same manner.

3.4 Magnet pole position.-The magnets shall be assembled into the body with the poles positioned to effect attraction between each pair.

3.5 Drop and plate penetration tests.-The charge shall detonate and create a hole of 9 square inches, minimum, in 1 inch thick boiler plate after being subjected to three 5 foot vertical drops.

3.6 Workmanship.-All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall free from burrs, chips, sharp edges, dirt, grease, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required marking shall be neat and sharply defined.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL-I-45208.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of analysis on all material procured directly by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL-I-45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Initial production inspection.

4.2.1 Submission.-The contractor shall submit an initial production sample consisting of (ten) consecutively produced assemblies and 10 complete sets of parts (i.e. 10 each of every individual component and every subassembly), which have been produced by the production process which the contractor will use in fulfilling the contract, to a Government-approved facility designated by the contracting officer. All parts and materials shall be obtained from the same source of supply as will be used in regular production. The initial production sample shall be accompanied by certificates of analyses for all materials. An initial production sample shall also be submitted prior to initiation of production and whenever a change occurs in manufacturing process, material used, drawing, specification, or source of supply such as to significantly effect product uniformity as determined by the Government.

4.2.2 Inspection to be performed.-Each assembly and component shall be inspected by the Government for all the inspection specified in 4.3.3.1 of this specification and for all requirements of the applicable drawings.

4.2.3 Rejection.-If any assembly or component fails to comply with the applicable requirements, the initial production sample shall be rejected and the contractor shall take the necessary corrective action and submit a new initial production sample or portion thereof, as specified by the contracting officer. The contractor shall continue to submit new or supplemental quantities as necessary until such time as the sample successfully passes the inspection specified or until otherwise directed by the contracting officer.

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4.3 Inspection provisions.

4.3.1 Lot formation.-A lot shall consist of charge assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, and same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit, as determined by the Government, shall not necessitate changing the lot number. Each lot shall contain:

a. Bodies produced under one set of pressing conditions from not more than one batch or homogeneous blend of batches of PBX.

4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4

4.3.2.1 Magnet (see dwg. 8826633 covering a detail of dwg. 8826645).

Categories	Defects	Method of Inspection	Code No. (see 6.3)
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Poles not flat or parallel	Gage	01001
102.	Distance from hole center to end of pole, minimum (min.)	Gage	01002
103.	Depth between poles, (min)	Gage	01003
Minor:	AQL 1.50 percent		
201.	Thickness, maximum (max.)	Gage	01004
202.	Distance between poles, min.	Gage	01005
203.	Evidence of poor workmanship (see 3.6)	Visual	01006

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4.3.2.2 Adapter, blasting cap (see dwg. 8826635 covering a detail of dwg. 8826645).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.25 percent		
101.	Pitch diameter of thread	Gage	02001
102.	Lip not flat	Visual	02002
Minor:	AQL 1.50 percent		
201.	Distance from lip to inside shoulder, min	Gage	02003
202.	Inside diameter of small end, min.	Gage	02004
203.	Evidence of poor workmanship (see 3.6)	Visual	02005

4.3.2.3 Keeper, magnet (see dwg. 8826641 covering a detail of dwg. 8826645).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 1.00 percent		
201.	Thickness, min	Gage	03001
202.	Evidence of poor workmanship (see 3.6)	Visual	03002

4.3.2.4 Body (see dwg. 8826643 covering a detail of dwg. 8826645).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Location of blasting cap hole	Gage	04001
102.	Location of pivot pin hole	Gage	04002
103.	Pivot pin hole not extending across cavity	Visual	04003

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Minor:	AQL 1.50 percent		
201.	Blasting cap hole not drilled entire length	Visual	04004
202.	Adapter lip countersink missing	Visual	04005
203.	Excessive chipping of edges	Visual	04006
204.	Evidence of poor workmanship (see 3.6)	Visual	04007

4.3.2.5 Assembly prior to paint, (see dwg. 8826645).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Magnet strength (see 3.3 and 4.4.2)	Test	05001
102.	Magnet missing	Visual	05002
103.	Cover missing, loose, or damaged	Visual	05003
104.	Magnet not free	Manual	05004
105.	Cover tacky	Manual	05005
106.	Adapter missing, damaged or loose	Visual-Manual	05006
107.	Closing plug frozen to adapter or missing	Visual-Manual	05007
108.	Magnets repel each other (see 3.4)	Visual-Manual	05008
109.	Block Warped (magnets out of alignment)	Visual	05009
Minor:	AQL 2.50 percent		
201.	Overall length, max	Gage	05010
202.	Overall width, max	Gage	05011
203.	Overall thickness, max	Gage	05012
204.	Magnet keeper missing	Visual	05013
205.	Block incorrectly wrapped	Visual	05014
206.	Evidence of poor workmanship (see 3.6)	Visual	05015
207.	Closing plug loose or damaged	Visual-Manual	05016

4.3.2.6 Assembly (see dwg. 8826645).

Categories	Defects	Method of Inspection	Code No.

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Critical: None defined

Major: None defined

Minor:		AQL 1.00 percent	
201.	Paint tacky	Manual	06001
202.	Marking missing, misleading or unidentifiable	Visual	06002

4.3.2.7 Carton, sealed (see dwg. 8798858).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.25 percent		
101.	Carton cut, split or damaged	Visual	07001
Minor:	AQL 2.50 percent		
201.	Label loose, torn or missing	Visual	07002
202.	Marking on label misleading or unidentifiable	Visual	07003
203.	Seal insecure	Visual	07004
204.	Corners not blunted	Visual	07005
205.	Contents loose	Manual	07006

4.3.2.8 Sealed barrier bag (see dwg. 8798858).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Incomplete seal	Visual	08001
102.	Sealed not permitting subsequent seals	Visual	08002
103.	Air not exhausted	Visual	08003
104.	Bag cut or damaged	Visual	08004
Minor:	AQL 1.00 percent		
201.	Label loose, torn or missing	Visual	08005
202.	Marking on label misleading or unidentifiable	Visual	08006

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4.3.2.9 Wood packing box, sealed (see dwg. 8798859).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Hardware missing, broken or loose.	Visual-Manual	09001
102.	Strapping missing, broken or loose	Visual-Manual	09002
103.	Contents exposed	Visual	09003
104.	Board broken or split	Visual	09004
105.	FSN number misleading or unidentifiable	Visual	09005
Minor:	AQL 2.50 percent		
201.	Hardware improperly engaged	Visual	09006
202.	Nail protruding	Visual	09007
203.	Strapping improperly assembled	Visual	09008
204.	Car seal missing, unsealed or improperly positioned	Visual	09009
205.	Marking misleading or unidentifiable	Visual	09010
206.	Contents loose	Manual	09011

4.3.3 Testing

4.3.3.1 Initial production sample. (see 4.2.1).-The samples submitted as the initial production quantity shall be tested as specified below after having been examined for all other drawing requirements. Rejection and resubmission shall be in accordance with 4.2.3.

4.3.3.1.1 Specific gravity of body. (see dwg. 8826643 and 4.4.1) - Code No. 04008.

4.3.3.1.2 Magnet strength (see 3.3 and 4.4.2) - Code No. 10001.

4.3.3.1.3 Drop test and functioning. (see 3.5 and 4.4.3) - Code No. 11001.

4.3.3.2 Production testing

4.3.3.2.1 Specific gravity of body (see dwg. 8826643) - Major

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defect - Code No. 04008.-A sample of 10 compressed blocks or bodies shall be selected from each lot for this determination which shall be performed as specified in 4.4.1. Failure of one or more units in meeting the requirement shall reject the lot.

4.3.3.2.2 Drop test and functioning (see 3.5) - Major defect - Code No. 11001.-A sample of 10 charge assemblies shall be selected from each lot for these tests which shall be performed as specified in 4.4.3. Failure of one or more units to create a hole equal to, or greater than, the requirement shall reject the lot.

4.3.4 Inspection equipment.-Index of Inspection Equipment Lists Number 8826645 identifies available approved inspection equipment designs. The contractor may, however, design alternate inspection equipment in order to facilitate his operation. Designs of any acceptance inspection equipment prepared by the contractor to support his approved written inspection plan (see 4.1.1) shall be approved by the Government prior to fabrication or procuring of such equipment (see 6.2). All acceptance inspection equipment shall be certified by the Government as complying with the approved design prior to its use. Supply and maintenance of inspection equipment shall be in accordance with Specification MIL-I-45607.

4.4 Test methods and procedures.

4.4.1 Specific gravity of body.-The body shall be weighed on a balance and the weight recorded. It shall then be suspended in water in such a manner that there can be no entrapped air and weighed again. Specific gravity shall be calculated as follows:

$$\text{Specific gravity} = \frac{A}{A-B}$$

where:

A = dry weight
B = weight in water

4.4.2 Magnet pull test.-A line shall be run through the blasting cap hole of the charge. The line shall be of sufficient length that when the ends are attached to each other and force is applied to the loop, the force will be equally applied against each magnet. The charge shall then be placed under the required flat plate and a 15 pound weight suspended on the loop. If the charge releases from the plate it shall be classed defective. If release does not occur, the charge shall be manually released from the plate and shall then be placed under the curved plate and an 8 pound weight shall be suspended

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from the loop. If release is effected, the charge shall be classed defective.

4.4.3 Drop and plate penetration tests.-The charge shall be assembled into a fixture which assures a guided, free drop onto a steel plate. Three separate drops shall be made, one each in the following positions: horizontal with the magnets down, vertical with an end down and angled so that a small edge adjacent to the magnets strikes the plate. After completion of the drop test the charge shall be detonated. The test plate shall have a clearance of a minimum of 4 inches from the side opposite the charge. (The charges may be initiated with Engineer Corps Electric Blasting Cap, Type JII.) For computing the area of the hole, the length and width shall be measured with a scale at what visually appears to be the mean dimensions. Any unit which fails to create a hole equal to, or greater than, the minimum requirement shall be classed defective. Subsequent units fired against a plate shall be so positioned that there is a minimum of 5 inches of unstressed metal in any direction.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level A.-The demolition charge shall be packaged and identified in accordance with Dwg. 8798858.

5.2 Packing and marking.

5.2.1 Level A.-The packaged assemblies shall be packed and marked in accordance with dwg. 8798859.

5.3 Data cards.-Data card information shall be as specified in Specification MIL-A-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number, and date of this specification.

6.2 Submission of contractor-developed equipment designs.-Documentation of contractor-developed inspection equipment designs shall be submitted in duplicate and identified with the associated examination or test required by the specification, to Picatinny Arsenal, Dover, New Jersey, ATTN: ORDBB-ND, for approval. This

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documentation shall include special equipment designs and a complete description of the use and application of all standard devices and set-ups as well as special equipment designs. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.3 Inspection code numbers.-The five-digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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